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Wang

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(54) **FOLDABLE SMALL FLYING AND SQUAT RACK**

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A63B 23/04 (2006.01)
A63B 23/12 (2006.01)

(52) **U.S. Cl.**
CPC ***A63B 21/154*** (2013.01); ***A63B 21/00069*** (2013.01); ***A63B 23/1254*** (2013.01); *A63B 2023/0411* (2013.01); *A63B 2210/50* (2013.01); *A63B 2225/10* (2013.01)

(58) **Field of Classification Search**
CPC A63B 21/154; A63B 21/00069; A63B 23/1254; A63B 2023/0411; A63B 2210/50; A63B 2225/10
See application file for complete search history.

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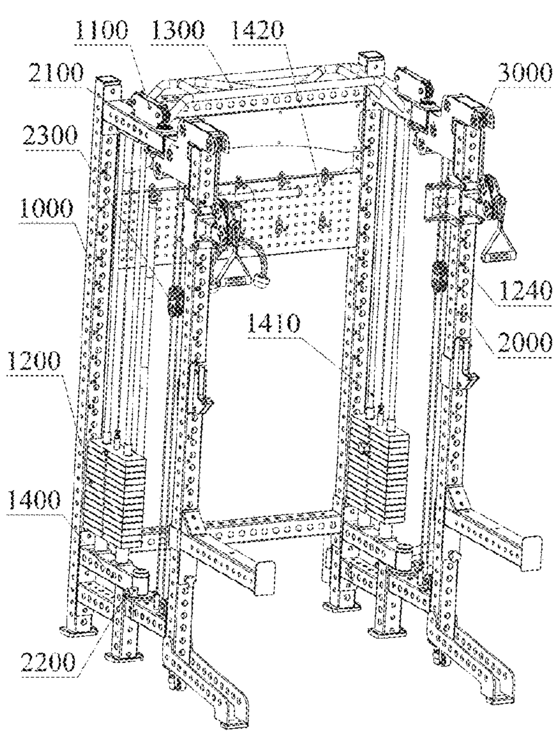
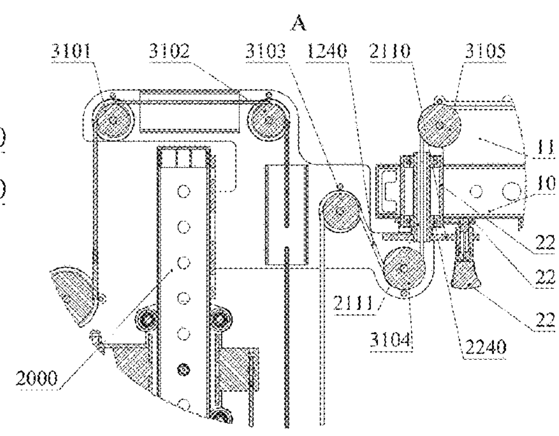
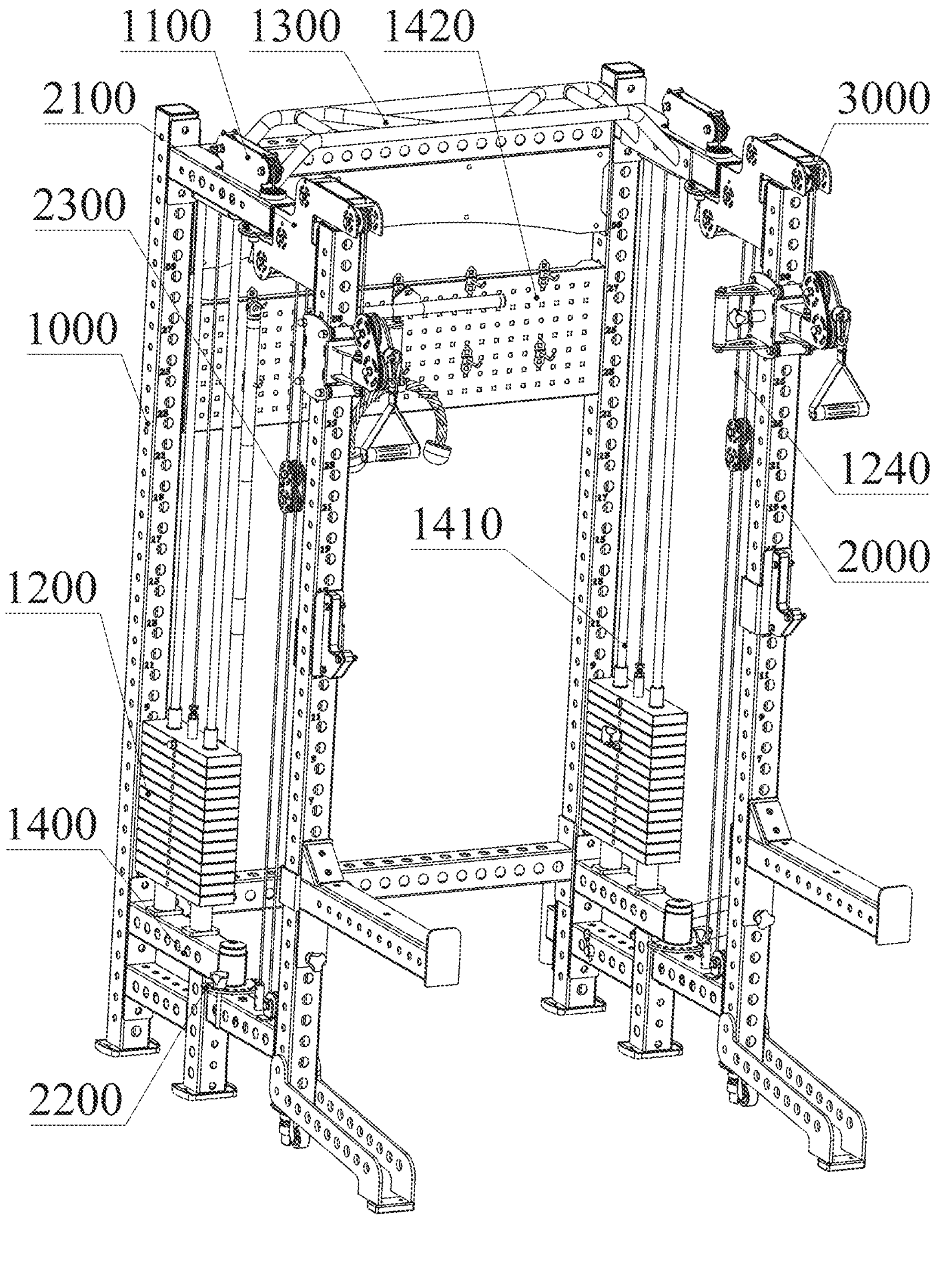
(57) **ABSTRACT**

The present disclosure provides a foldable small flying and squat rack, which comprises a fixed rack, a pivot rack and pulley blocks disposed on the fixed rack and the pivot rack, wherein the pivot rack can rotate relative to the fixed rack, a counterweight assembly is disposed at the lower part of the fixed rack, and a connecting rope is connected at the upper part of the counterweight assembly; the connecting rope winds around the pulley blocks on the fixed rack and the pivot rack in turn to extend out, and then passes through a flying guide wheel to be connected to a grip part.

20 Claims, 9 Drawing Sheets







1100
1300
1420
2100
3000
2300
1000
1240
1410
2000
1200
1400
2200

FIG. 1

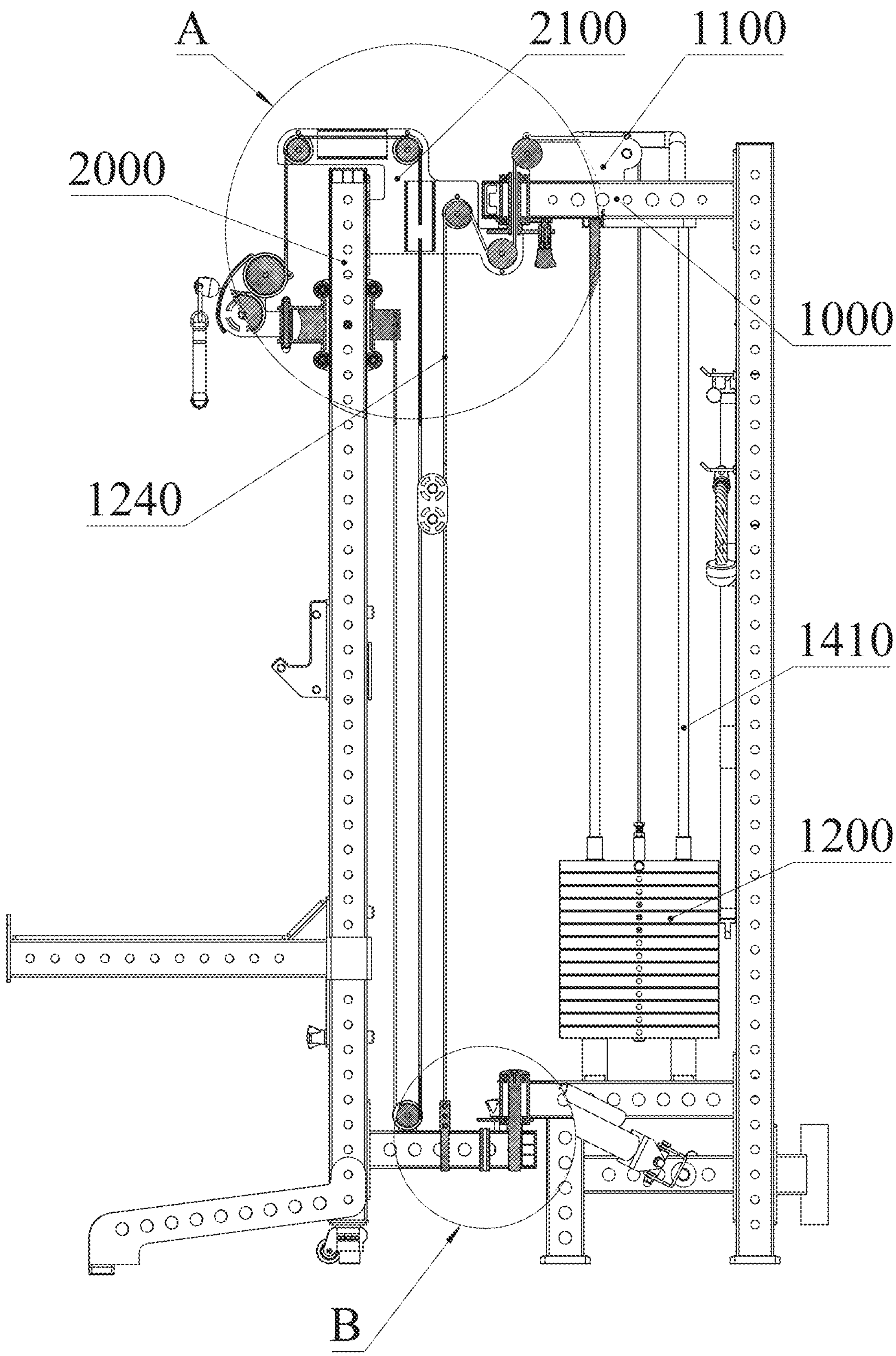


FIG. 2

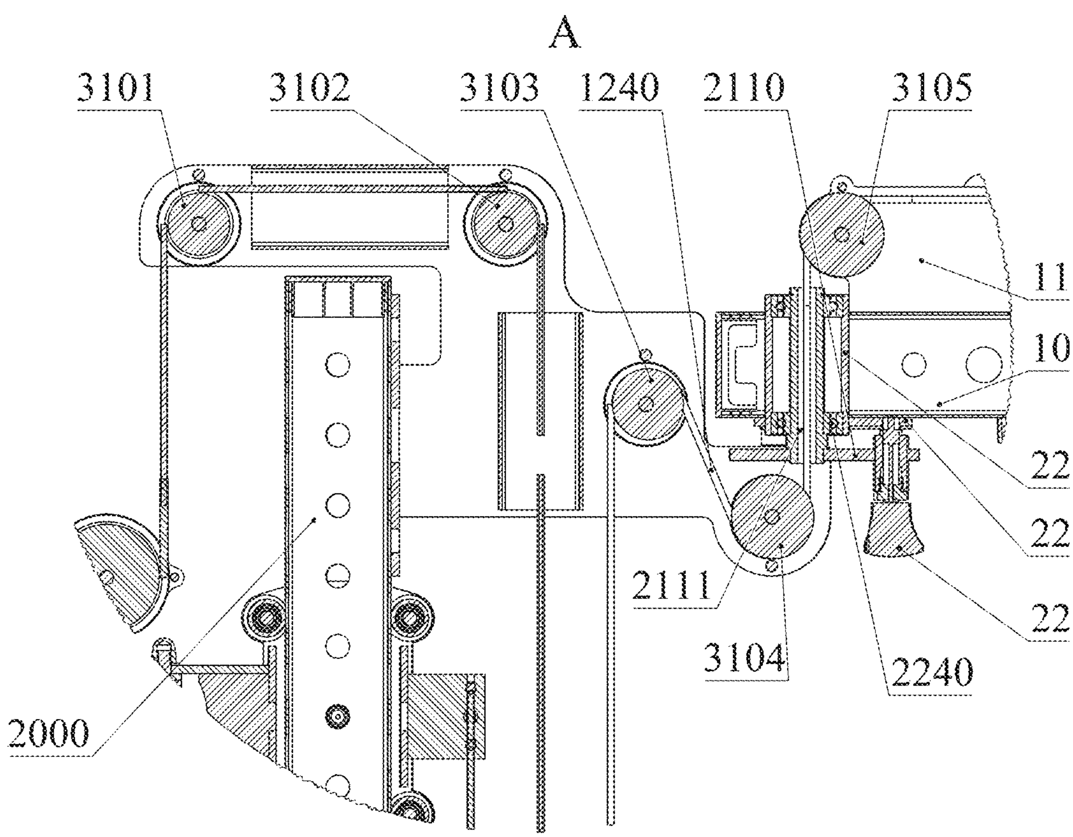

A
3101
3102
3103
1240
2110
3105
11
10
22
22
22
2111
3104
2240
2000

FIG. 3

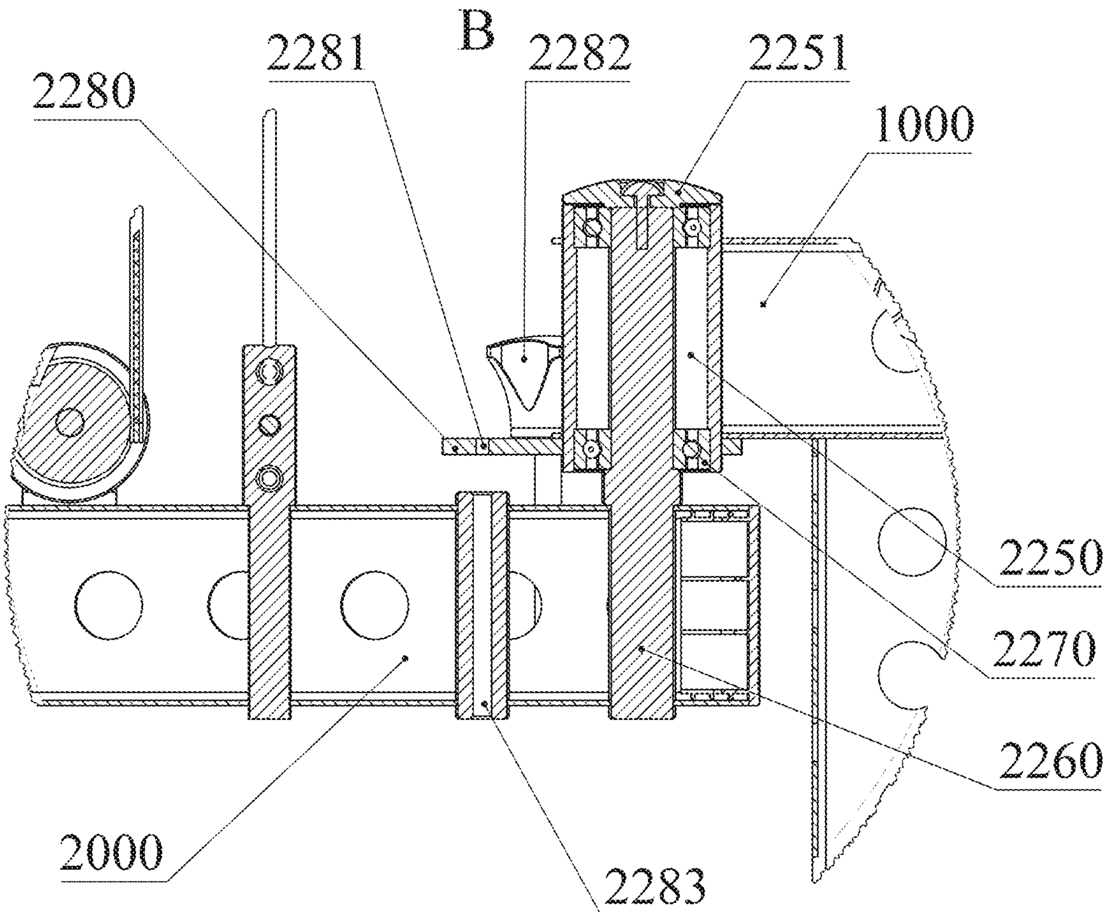


FIG. 4

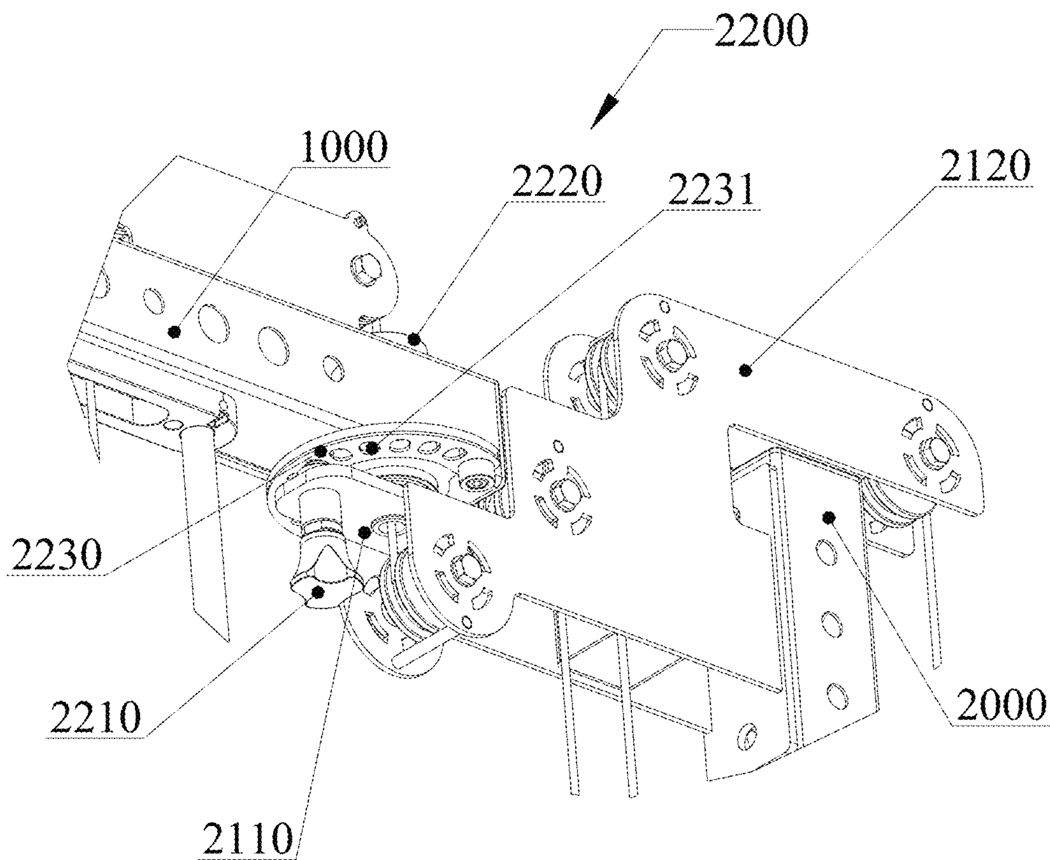


FIG. 5

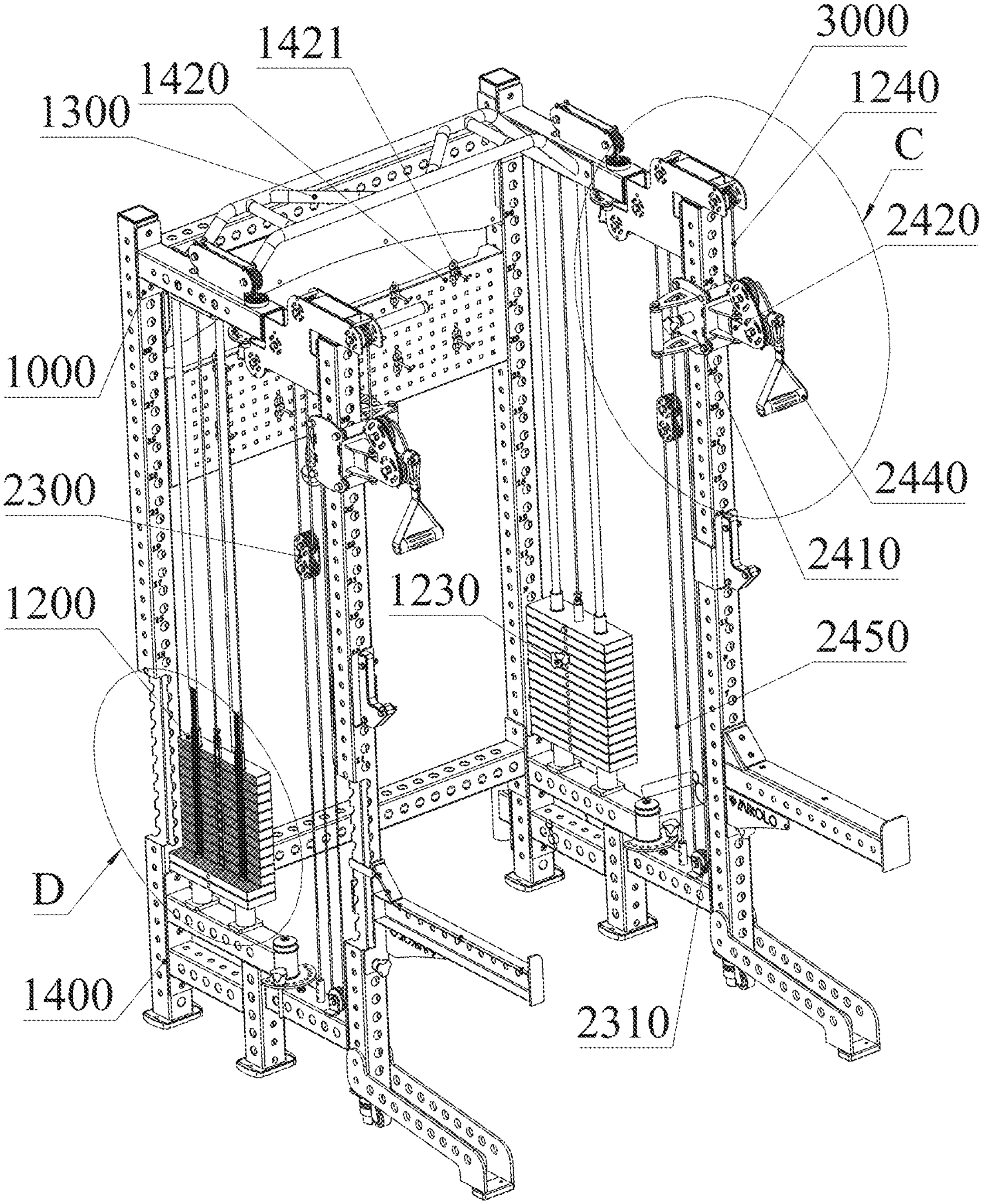

3000
1421
1420
1300
1240
C
2420
1000
2440
2300
2410
1230
1200
2450
D
1400
2310

FIG. 6

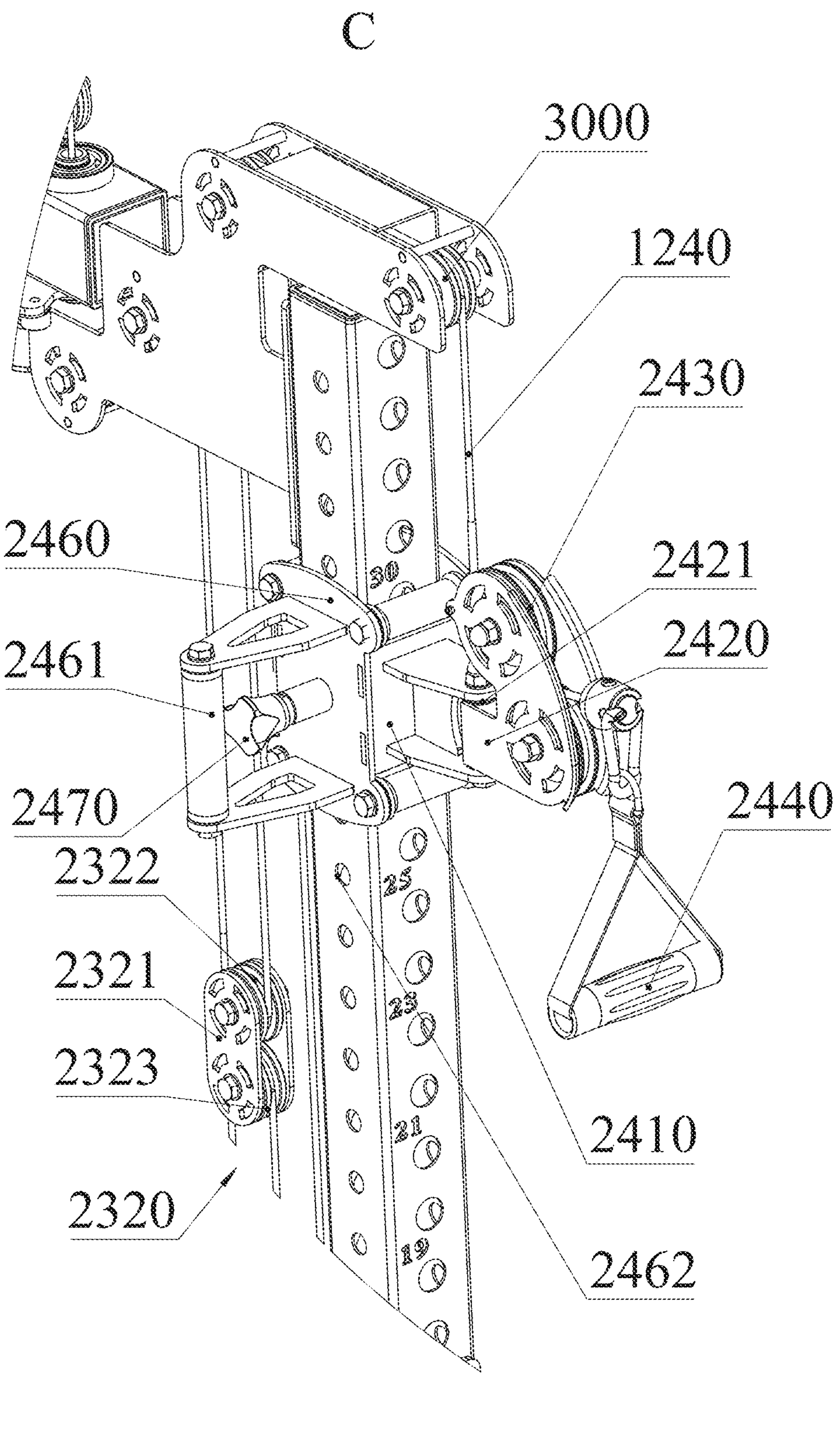

C
3000
1240
2430
2460
2421
2420
2461
2470
2440
2322
2321
2323
2320
2410
2462
30
25
23
21
19

FIG. 7

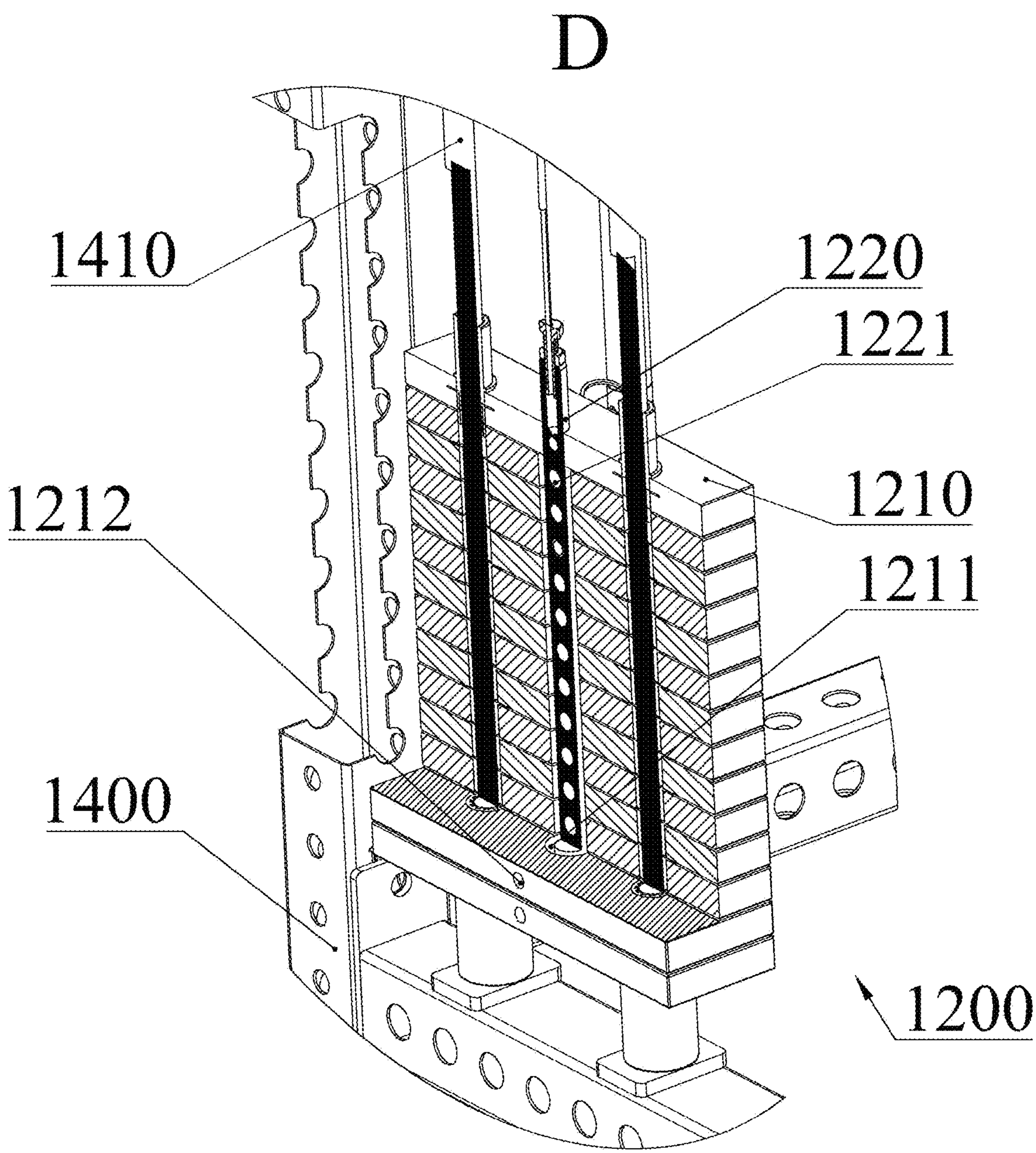


FIG. 8

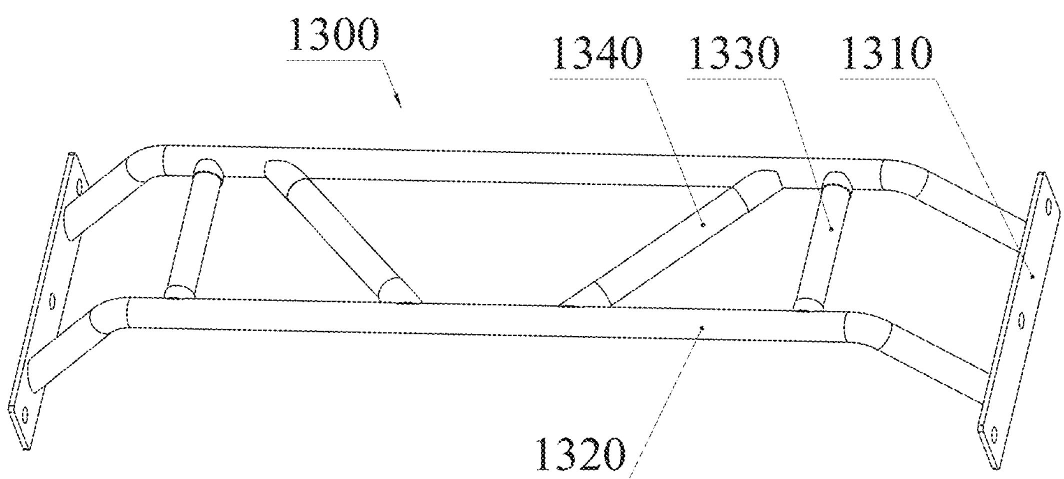


FIG. 9

FOLDABLE SMALL FLYING AND SQUAT RACK

TECHNICAL FIELD

The present disclosure relates to the technical field of sports equipment, in particular to a foldable small flying and squat rack.

BACKGROUND

Smith machine and big and small flying exercisers are two kinds of fitness equipment that must be equipped in the exercise field and gym. However, having these two exercisers at the same time is too expensive and takes up too much space, which is not suitable for use in places with limited space such as home.

Although there are comprehensive exercisers combining Smith machine and small flying exerciser in the market, Smith machine and small flying exerciser are simply superimposed and combined together, and they do not have the function of mutual conversion. For this kind of machines, since they include all the necessary components of Smith machine and small flying exerciser, the manufacturing cost is not less than that of two separate machines, and they are huge.

A "Slidable bar and carriage exercise assembly" is known in the art, which includes a Smith bar as a part of multi-functional exerciser, and uses Smith bar instead of free weight to use optional weight stacking to resist movement resistance; U.S. patent Ser. No. 12/011,635 puts forward "Multifunctional comprehensive training device capable of being grasped stably", which enables the same set of selected barbells to be used for Smith mechanism and cable mechanism by racks, hangers and other parts.

However, when the exerciser disclosed above is adjusted in angle and folded for storage, the rotation of the pivot rack often involves the pull rope and then pulls the load-bearing assembly, which increases the rotation resistance of the pivot rack.

SUMMARY

In order to solve the shortcomings of the prior art, the present disclosure provides a foldable small flying and squat rack, which includes a fixed rack, pivot racks and pulley blocks disposed on the fixed rack and the pivot rack, wherein there two pivot racks that are respectively rotatably connected to the fixed rack; and

- a lower part of the fixed rack is provided with a counterweight assembly, which is connected with the fixed rack in a sliding manner, and an upper part of the counterweight assembly is connected with a connecting rope, which winds around the pulley blocks on the fixed rack and the pivot racks in turn to extend out; and
- each of the two pivot racks is provided with a pivot wheel set mechanism, and a connecting sheet is fixedly disposed on the pivot wheel set mechanism; and a rotary inner cylinder is fixedly disposed on the connecting sheet, and the pivot racks are rotatably connected with the fixed rack through the rotary inner cylinder; and the connecting rope passes through the rotary inner cylinder, and the pulley block includes a first deflecting pulley and a second deflecting pulley, which are respectively disposed above and below the rotary inner cylinder; and
- wherein, a part of the connecting rope located at the position of a rotating axis of the rotary inner cylinder is always in a vertical state, and a tangent point of the connecting rope winding out of the first deflecting pulley and into the second deflecting pulley is located on the axis of the rotary inner cylinder, so as to avoid the connecting rope from being dragged by the rotation of the pivot rack.

At the same time, the present disclosure provides a foldable small flying and squat rack, which includes a fixed rack, pivot racks and pulley blocks disposed on the fixed rack and the pivot rack, wherein there are two pivot racks that are respectively connected to the fixed rack; and

- a lower part of the fixed rack is provided with a counterweight assembly, which is connected with the fixed rack in a sliding manner, and an upper part of the counterweight assembly is connected with a connecting rope, which winds around the pulley blocks on the fixed rack and the pivot racks in turn to extend out; and
- each of the two pivot racks is provided with a pivot wheel set mechanism, and a connecting sheet is fixedly disposed on the pivot wheel set mechanism; and a rotary inner cylinder is fixedly disposed on the connecting sheet, the two pivot racks are rotatably connected with the fixed rack respectively through the rotary inner cylinder and rotate to any angle within a limited angle range; and
- the connecting rope passes through the rotary inner cylinder, and the pulley block includes a first deflecting pulley and a second deflecting pulley, which are respectively disposed above and below the rotary inner cylinder; and
- wherein, a part of the connecting rope located at the position of a rotating axis of the rotary inner cylinder is always in a vertical state, and a tangent point of the connecting rope winding out of the first deflecting pulley and into the second deflecting pulley is located on the axis of the rotary inner cylinder, so as to avoid the connecting rope from being dragged by the rotation of the pivot rack.

At the same time, the present disclosure also provides a foldable small flying and squat rack, which includes a fixed rack, pivot racks and pulley blocks disposed on the fixed rack and the pivot rack, wherein there are two pivot racks that are respectively connected to the fixed rack; and

- a lower part of the fixed rack is provided with a counterweight assembly, which is connected with the fixed rack in a sliding manner, and an upper part of the counterweight assembly is connected with a connecting rope, which winds around the pulley blocks on the fixed rack and the pivot racks in turn to extend out; and
- each of the two pivot racks is provided with a pivot wheel set mechanism, and a connecting sheet is fixedly disposed on the pivot wheel set mechanism; and a rotary inner cylinder is fixedly disposed on the connecting sheet, the two pivot racks are rotatably connected with the fixed rack respectively through the rotary inner cylinder; and
- the pulley block includes a first deflecting pulley and a second deflecting pulley; the first deflecting pulley is disposed above the rotary inner cylinder, and the second deflecting pulley is disposed below the rotary inner cylinder;
- wherein, the connecting rope winds out of the first deflecting pulley, passes through the rotary inner cylinder and winds into the second deflecting pulley; and the connecting rope in the rotary inner cylinder is always kept

in a vertical state, so as to avoid the connecting rope from being dragged by the rotation of the pivot rack.

BRIEF DESCRIPTION OF DRAWINGS

In order to explain the technical solution of this application more clearly, the drawings needed in the implementation will be briefly introduced below. Obviously, the drawings described below are only some implementations of this application. For those skilled in the art, other drawings can be obtained according to these drawings without creative work.

FIG. **1** is a first perspective schematic view of a foldable small flying and a squat rack;

FIG. **2** is a schematic right view of the foldable small flying and squat rack;

FIG. **3** is a partial cross-sectional view at A of FIG. **2**;

FIG. **4** is a partial cross-sectional view at B in FIG. **2**;

FIG. **5** is a partial schematic view of the hole plate fixing mechanism;

FIG. **6** is a second perspective view of the foldable small flying and squat rack;

FIG. **7** is a partial schematic view at C in FIG. **6**;

FIG. **8** is a partial sectional view at D in FIG. **6**;

FIG. **9** is a partial schematic view of the pull-up assembly.

REFERENCE SIGNS

1000, Fixed rack; **1100**, Fixed wheel carrier; **1200**, Counterweight assembly; **1210**, Weight stack; **1211**, Yielding hole; **1212**, Adjusting hole; **1220**, Adjusting rod; **1221**, Matching hole; **1230**, Adjusting pin; **1240**, Connecting rope; **1300**, Pull-up assembly; **1310**, Fixed plate; **1320**, Fixed tube; **1330**, Vertical rod; **1340**, Inclined rod; **1400**, Chassis; **1410**, Counterweight slidable bar; **1420**, Storage plate; **1421**, Hook; **2000**, Pivot rack; **2100**, Pivot wheel set mechanism; **2110**, Connecting sheet; **2111**, Rotary inner cylinder; **2120**, Pivot wheel rack; **2200**, Hole plate fixing mechanism; **2210**, First spring bolt; **2220**, Upper rotary sleeve; **2230**, Upper fixed hole plate; **2231**, Upper insertion hole; **2240**, Upper pivot bearing; **2250**, Lower rotary sleeve; **2251**, circular cover plate; **2260**, Rotary column; **2270**, Lower pivot bearing; **2280**, Lower fixed hole plate; **2281**, Lower insertion hole; **2282**, Fixed pin; **2283**, Positioning hole; **2300**, Adjusting wheel set mechanism; **2310**, Crown block; **2320**, Movable pulley block; **2321**, Sliding rack; **2322**, Upper sliding wheel; **2323**, Lower sliding wheel; **2410**, Flying carriage; **2420**, Flying wheel carrier; **2421**, Wheel carrier shaft; **2430**, Flying guide wheel; **2440**, Grip part; **2450**, Adjusting rope; **2460**, Carriage rack; **2461**, Handle; **2462**, Preset hole; **2470**, Second spring bolt; **3000**, Pulley block; **3101**, First pivot wheel; **3102**, Second pivot wheel; **3103**, Third pivot wheel; **3104**, Fourth pivot wheel; **3105**, Fixed wheel.

DESCRIPTION OF EMBODIMENTS

The technical solution in the embodiment of the application will be clearly and completely described below with reference to the drawings in the embodiment of the application. Obviously, the described embodiment is only part of, rather than all of the embodiments of the application. Based on the embodiments in this application, all other embodiments obtained by those skilled in the art without creative labor belong to the protection scope of this application.

Reference to "an example" or "an embodiment" herein means that a particular feature, structure or characteristic described in connection with an example or an embodiment can be included in at least one embodiment of this application. The appearance of this phrase in various places in the specification does not necessarily refer to the same embodiment, nor is it an independent or alternative embodiment mutually exclusive with other embodiments. It is understood explicitly and implicitly by those skilled in the art that the embodiments described herein can be combined with other embodiments.

In this specification, for the sake of convenience, words and expressions indicating orientation or positional relationship such as "middle", "upper", "lower", "front", "rear", "vertical", "horizontal", "top", "inner" and "outer" are used to illustrate the positional relationship of constituent elements with reference to the attached drawings, which is only for the convenience of describing this specification and simplifying the description, and does not indicate or imply that the referred devices or elements must have a specific orientation, be constructed and operated in a specific orientation, so it cannot be understood as a limitation of this disclosure. The positional relationship of the constituent elements is appropriately changed according to the direction of the described constituent elements. Therefore, it is not limited to the words and expressions described in the specification, and can be replaced appropriately according to the situation.

As shown in FIGS. **1** and **2**, the present disclosure provides a foldable small flying and squat rack, which includes a fixed rack **1000**, two pivot racks **2000**, and pulley blocks **3000** disposed on the fixed rack **1000** and the pivot racks **2000**.

Specifically, the pivot racks **2000** are rotatably connected to both sides of the fixed rack **1000**, and the pivot racks **2000** are symmetrically disposed about the fixed rack **1000**. In order to meet the diversified training needs of users, a counterweight assembly **1200** is disposed at the lower part of the fixed rack **1000**, and a connecting rope **1240** is connected at the upper part of the counterweight assembly **1200**, which sequentially winds around pulley blocks **3000** on the fixed rack **1000** and the pivot racks **2000** to extend out to ensure that the connecting rope **1240** transmits the lifting force to the counterweight assembly **1200**.

As shown in FIG. **2** and FIG. **3**, the pivot rack **2000** is provided with a pivot wheel set mechanism **2100**, and a connecting sheet **2110** is fixedly disposed on the pivot wheel set mechanism **2100**. A rotary inner cylinder **2111** is fixedly disposed on the connecting sheet **2110**, so that the pivot rack **2000** can be rotationally connected with the fixed rack **1000** by the rotary inner cylinder **2111**. The connecting rope **1240** passes through the rotary inner cylinder **2111**, and the pivot rack **2000** can horizontally rotate to any position within a limited angle range to rotate to a normal open state, a storage state or an expanded state.

In order to ensure that the connecting rope **1240** is not dragged by the rotation of the pivot rack **2000**, the pulley block **3000** is provided with a first deflecting pulley and a second deflecting pulley, which are respectively disposed above and below the rotary inner cylinder **2111**. Furthermore, when the connecting rope **1240** winds out of the first deflecting pulley and the second deflecting pulley, it is preferable that the connecting rope **1240** coincides with the rotating axis of the rotary inner cylinder **2111**. Even if the pivot rack **2000** rotates around the rotary inner cylinder **2111**, the part of the connecting rope **1240** located at the rotating axis of the rotary inner cylinder **2111** always remains vertical. Thus, although the connecting rope **1240** has torsional shear stress between the first deflecting pulley and the second deflecting pulley, it does not affect the lifting force on the counterweight assembly **1200** connected at the end of the connecting rope **1240**, which ensures that when the connecting rope **1240** is in the abnormal opening state of the pivot racks **2000**, the counterweight assembly **1200** is lifted due to the pulling force. In other embodiments of the connecting rope **1240** (not shown), the connecting rope **1240** does not need to coincide with the axis of the rotary inner cylinder **2111**, but only needs to be parallel to the axis of the rotary inner cylinder **2111**, and the connecting rope **1240** does not touch the inner wall of the rotary inner cylinder **2111**, which means that the connecting rope **1240** is offset by a certain distance relative to the axis of the rotary inner cylinder **2111**, thus ensuring the normal operation of the connecting rope **1240**.

In order to ensure the transmission of the lifting force of the connecting rope **1240**, the pivot wheel set mechanism **2100** further includes a pivot wheel rack **2120** fixedly connected to the top of the pivot rack **2000**. The first pivot wheel **3101**, the second pivot wheel **3102**, the third pivot wheel **3103** and the fourth pivot wheel **3104** in the pulley block **3000** are disposed on the pivot wheel rack **2120**. The first pivot wheel **3101** is located at the side of the pivot rack **2000** far from the fixed rack **1000**, while the second pivot wheel **3102**, the third pivot wheel **3103** and the fourth pivot wheel **3104** are located at one side of the pivot rack **2000** near the fixed rack **1000**. The connecting rope **1240** passes through the fourth pivot wheel **3104**, the third pivot wheel **3103**, the second pivot wheel **3102** and the first pivot wheel **3101** in turn, wherein the fourth pivot wheel **3104** is disposed below the rotary inner cylinder **2111** to constitute the second deflecting pulley.

As shown in FIG. **3**, both sides of the upper part of the fixed rack **1000** are provided with fixed wheel carriers **1100**, each fixed wheel carrier **1100** is rotatably connected with two fixed wheels **3105**, and the fixed wheel **3105** close to the pivot rack **2000** is disposed above the rotary inner cylinder **2111** to constitute the first deflecting pulley.

As shown in FIG. **4** and FIG. **5**, in order to provide a selectin for the rotation angle of the pivot rack **2000**, a hole plate fixing mechanism **2200** for limiting the rotation of the pivot rack **2000** is disposed between the pivot rack **2000** and the fixed rack **1000**. A first spring bolt **2210** is also disposed on the connecting sheet **2110**. The hole plate fixing mechanism **2200** includes an upper rotary sleeve **2220** fixedly connected with the upper part of the fixed rack **1000**. The rotary inner cylinder **2111** and the upper rotary sleeve **2220** are rotatably connected, and the upper rotary sleeve **2220** is fixedly connected with an upper fixed hole plate **2230**, which is provided with a plurality of upper insertion holes **2231**, wherein the first spring pin **2210** is inserted into the upper insertion hole **2231** to fix the position of the connecting sheet **2110** relative to the upper fixed hole plate **2230**, and the axes of the upper rotary sleeve **2220** and the rotary inner cylinder **2111** are perpendicular to the ground. A plurality of upper pivot bearings **2240** are disposed between the upper rotary sleeve **2220** and the rotary inner cylinder **2111**. The upper rotary sleeve **2220** and the rotary inner cylinder **2111** are hollow, and the connecting rope **1240** passes through the rotary inner cylinder **2111**. When the angle of the pivot rack **2000** needs to be changed, the user can partially pull out the first spring bolt **2210** first, so that the first spring bolt **2210** is separated from the upper insertion hole **2231**. The user rotates the pivot rack **2000** to a proper position, and then loosens his grip, and the first spring bolt **2210** will automatically rebound to be inserted into the insertion hole **2231** on the upper fixed hole plate **2230** to achieve locking.

As shown in FIG. **4**, the hole plate fixing mechanism **2200** further includes a lower rotary sleeve **2250** fixedly connected with the lower part of the fixed rack **1000**, a rotary column **2260** is disposed below the pivot rack **2000**, and a plurality of lower pivot bearings **2270** are disposed between the lower rotary sleeve **2250** and the rotary column **2260**. In order to ensure that the pivot rack **2000** is firmly fixed, the rotating shafts of the upper rotary sleeve **2220** and the lower rotary sleeve **2250** coincide and are perpendicular to the ground.

As shown in FIG. **4**, in order to further ensure the stable rotation and firm fixation of the pivot rack **2000**, a lower fixed hole plate **2280** is fixedly disposed below the lower rotary sleeve **2250**, and a plurality of lower insertion holes **2281** are disposed on the lower fixed hole plate **2280**. A positioning hole **2283** is disposed on the pivot rack **2000**, and a fixed pin **2282** is inserted into the lower insertion hole **2281**. The fixed pin **2282** can be inserted into the positioning hole **2283** to fix the position of the lower fixed hole plate **2280** relative to the pivot rack **2000**. The top of the lower rotary sleeve **2250** is provided with a circular cover plate **2251**, which is fixedly connected with the rotary column **2260** by bolts. The use method of the fixed pin **2282** is the same as that of the first spring bolt **2210**, and they jointly play the role of fixing the pivot rack **2000**.

As shown in FIGS. **6** and **7**, both pivot racks **2000** are provided with adjusting wheel set mechanisms **2300**. The adjusting wheel set mechanism **2300** includes a crown block **2310** and a movable pulley set **2320**. The crown block **2310** is rotatably connected with the bottom of the pivot rack **2000**, and the movable pulley set **2320** is located in the middle of the pivot rack **2000**. The movable pulley block **2320** includes a sliding rack **2321**, and an upper sliding wheel **2322** and a lower sliding wheel **2323** rotatably connected with the sliding rack **2321**. The adjusting wheel set mechanism **2300** can cooperate with the flying mechanism to realize the up-and-down adjustment of the flying mechanism on the pivot rack **2000** to adapt to different users.

As shown in FIG. **7**, in order to make users use the flying mechanism easily, two pivot racks **2000** are both provided with flying carriages **2410**, which are movably connected with the pivot rack **2000**. The two bird carriages **2410** are both provided with flying wheel carriers **2420**. Two flying guide wheels **2430** are rotatably connected in the flying wheel carrier **2420**, and the flying wheel carrier **2420** is rotatably connected with the flying carriage **2410**. One end of the connecting rope **1240** is connected to the flying carrier **2420**. The connecting rope **1240** passes through the first pivot wheel **3101**, the second pivot wheel **3102**, the upper sliding wheel **2322**, the third pivot wheel **3103** and the fourth pivot wheel **3104** from the flying wheel carrier **2420**, passes through the hole plate fixing mechanism **2200**, and then passes through the fixed wheel carrier **1100**, and is connected with the counterweight assembly **1200**. The connecting rope **1240** is always in a vertical state at the position of the hole plate fixing mechanism **2200**, and the vertical line where the connecting rope **1240** is located is collinear with the rotating axis of the pivot rack **2000**, so that the user can adjust the opening angle of the pivot rack **2000** according to his own conditions, so as to adjust the use width between the two flying carriages **2410** and achieve more suitable exercise.

Furthermore, the position of the flying carriage **2410** is provided with a grip part **2440** to improve the user's comfort when exercising. The grip part **2440** is connected with the counterweight assembly **1200** through the connecting rope **1240**. The flying carriage **2410** is further provided with an adjusting rope **2450**. One end of the adjusting rope **2450** is connected with the flying carriage **2410**, passes through the crown block **2310** and the movable pulley block **2320** in turn, and the other end is connected with the bottom of the pivot rack **2000**.

The flying carriage **2410** includes a carriage rack **2460** slidably connected with the pivot rack **2000**, and two rollers are rotatably connected to both sides of the pivot rack **2000** in the carriage rack **2460**, wherein one side of the carriage rack **2460** is provided with a handle **2461**, and a second spring bolt **2470** is provided on the side of the carriage rack **2460** where the handle **2461** is provided. The pivot rack **2000** is provided with a plurality of transverse preset holes **2462**. The second spring pin **2470** can be inserted into the preset hole **2462** for fixation. When the user needs to adjust the height of the flying mechanism, he can first hold the handle **2461** and pull out the second spring pin **2470**, then slide the flying carriage **2410** to a proper position through the handle **2461** to be aligned with the preset hole **2462** on the pivot rack **2000**, and then allow the second spring pin **2470** to pass through the flying carriage **2410** to be inserted into the preset hole **2462**.

As a further illustration, because the preset holes **2462** in the pivot racks **2000** on both sides are horizontally disposed, the flying carriages **2410** at both ends can be ensured to be at the same horizontal position, and because the flying wheel carrier **2420** is rotatably connected with the flying carriage **2410** through the wheel carrier shaft **2421**, when the user uses the flying mechanism to exercise, the wheel carrier shaft **2421** will force the flying wheel carrier **2420** to rotate to a proper position due to the pulling of the user. The connecting rope **1240** at the grip part **2440** can transfer traction more smoothly, increase the fit with the flying guide wheel **2430**, reduce the fatigue damage of the equipment, and improve the service life of the equipment. At the same time, when the user adjusts the flying carriage **2410** to ascend or descend for a certain distance, the adjusting rope **2450** at the side of the crown block **2310** also ascends or descends for a certain distance, and the adjusting rope **2450** at the side of the movable pulley block **2320** ascends or descends by half. The connecting rope **1240**, which is also located on the side of the movable pulley set **2320**, also moves up or down by a half distance. Because the pulley is a movable pulley, the connecting rope **1240** will move twice the moving distance of the movable pulley set **2320**, thus ensuring that the grip part **2440** always stays at the fixed position of the flying carriage **2410** no matter how much the flying mechanism moves up or down.

As shown in FIG. **9**, in order to facilitate the user to do pull-up exercises, a pull-up assembly **1300** is disposed at the upper part of the fixed rack **1000**. The pull-up assembly **1300** includes two fixed plates **1310** fixedly connected with the fixed rack **1000**, wherein through holes can be uniformly disposed on the fixed plates **1310**, two cross beams are disposed at the upper part of the chassis **1400** of the fixed rack **1000**, and the pull-up assembly **1300** is erected between the cross beams. The fixed plate **1310** is closely attached to the cross beam, and the pull-up assembly **1300** is firmly connected to the fixed rack **1000** through the matching of the transverse preset hole **2462** on the cross beam and the through hole on the fixed plate **1310**. The cross beam has enough length to install the pull-up assembly **1300**, so that even if the pivot rack **2000** rotates, it is not necessary to disassemble the pull-up assembly **1300**, and users can perform pull-up exercises at any time. Two fixed tubes **1320** are fixedly connected between the two fixed plates **1310**, and two vertical rods **1330** and two inclined rods **1340** are fixedly connected between the two fixed tubes **1320**. All parts of the pull-up assembly **1300** are firmly connected, and the connection mode can be, but is not limited to, welding, riveting, threaded connection and the like. The vertical rods **1330** and the inclined rods **1340** can not only strengthen the firmness of the pull-up assembly **1300**, but also serve as exercise parts for pull-ups.

As shown in FIG. **8**, the counterweight assembly **1200** includes a plurality of weight stacks **1210**, and the middle of each weight stack **1210** is provided with a yielding hole **1211** and an adjusting hole **1212**. The yielding hole **1211** is disposed at the center of the horizontal plane of the weight stack **1210** and the adjusting hole **1212** is disposed at the center of the lateral surface of the weight stack **1210**. Both the yielding hole **1211** and the adjusting hole **1212** penetrate through the weight stacks **1210** and intersect at the center of the weight stacks **1210**. One end of the connecting rope **1240** close to the counterweight assembly **1200** is connected with an adjusting rod **1220**, which is inserted into the yielding hole **1211**. The adjusting rod **1220** is provided with a plurality of matching holes **1221**. When the adjusting rod **1220** is inserted into the yielding hole **1211**, the axes of the matching holes **1221** are respectively aligned with the axes of the corresponding adjusting holes **1212**. Adjusting pins **1230** are inserted into the adjusting holes **1212**. The adjusting pins **1230** can pass through both the adjusting holes **1212** and the matching holes **1221**, and the adjusting rod **1220** can lift all the weight stacks **1210** at the positions where the adjusting pins **1230** are inserted into the adjusting hole **1212** and those at the positions above, thus achieving the function of adjusting the weight.

As shown in FIG. **6**, the fixed rack **1000** includes a chassis **1400**, a counterweight slidable bar **1410** and a storage plate **1420** disposed on the chassis **1400**. The counterweight assembly **1200** is slidably connected with the counterweight slidable bar **1410**, and the counterweight slidable bar **1410** is parallel to the connecting rope **1240** above the adjusting rod **1220**. The storage plate **1420** is provided with a plurality of hooks **1421**, which can be used to hang various exercise parts.

The foldable small flying and squat rack provided by the present disclosure takes into account the common exercise function of both a flying exercise machine and a comprehensive squat rack, and includes a pivot rack **2000** which can rotate relative to the fixed rack **1000**. The pivot rack **2000** can be adjusted in different working states according to the needs of users, so that it is always in the desired state of users. At the same time, when not in use, the pivot rack **2000** can be adjusted to a storage state to reduce its occupied volume. The fixed rack **1000** is also provided with optional weight stacks **1210**. Optional weight stacks **1210** are provided by a designed connecting rope **1240**, so that the flying machine is adapted to users with different arm lengths by a designed adjusting rope **2450**. At the same time, the pull-up assembly **1300** is fixed above the fixed rack **1000** to facilitate customers to exercise for different parts. Moreover, the whole frame of the foldable small flying and squat rack is provided with preset holes **2462**, so that the height of the flying mechanism can be adjusted to adapt to different users.

The technical means disclosed in the solution of the present disclosure are not limited to the technical means disclosed in the above embodiments, but also include the technical solution composed of any combination of the above technical features. It should be appreciated that for those skilled in the art, several improvements and embellishments can be made without departing from the principle of the present disclosure, and these improvements and embellishments shall be regarded as falling within the protection scope of the present disclosure.

What is claimed is:

1. A foldable small flying and squat rack, comprising
a fixed rack, pivot racks and pulley blocks disposed on the fixed rack and the pivot racks, wherein the pivot racks comprise two pivot racks that are respectively rotatably connected to the fixed rack; and
a counterweight assembly connected with the fixed rack in a sliding manner, wherein an upper part of the counterweight assembly is connected with a connecting rope-, which winds around the pulley blocks on the fixed rack and the pivot racks in turn to extend out;
wherein each of the two pivot racks is provided with a pivot wheel set mechanism, and a connecting sheet is fixedly disposed on the pivot wheel set mechanism; and a rotary inner cylinder is fixedly disposed on the connecting sheet, and the pivot racks are rotatably connected with the fixed rack through the rotary inner cylinder; and the connecting rope passes through the rotary inner cylinder, and a pulley block of the pulley blocks comprises a first deflecting pulley and a second deflecting pulley, which are respectively disposed above and below the rotary inner cylinder; and
wherein, a part of the connecting rope located at the position of a rotating axis of the rotary inner cylinder is always in a vertical state, and a tangent point of the connecting rope winding out of the first deflecting pulley and into the second deflecting pulley is located on the axis of the rotary inner cylinder, so as to avoid the connecting rope from being dragged by the rotation of a pivot rack of the pivot racks.

2. The foldable small flying and squat rack according to claim 1, wherein the pivot wheel set mechanism comprises a pivot wheel rack fixedly connected to the top of the pivot rack, and a pulley block of the pulley blocks comprises a first pivot wheel, a second pivot wheel, a third pivot wheel and a fourth pivot wheel rotatably connected to the pivot wheel rack;
the first pivot wheel is located on a side of the pivot rack far from the fixed rack, and the second pivot wheel, the third pivot wheel and the fourth pivot wheel are located on a side of the pivot rack close to the fixed rack; and the connecting rope passes through the fourth pivot wheel, the third pivot wheel, the second pivot wheel and the first pivot wheel in turn; and
wherein the fourth pivot wheel is disposed below the rotary inner cylinder to constitute the second deflecting pulley.

3. The foldable small flying and squat rack according to claim 2, wherein fixed wheel carriers are disposed at both sides of an upper part of the fixed rack, and each fixed wheel carrier is rotatably connected with two fixed wheels, wherein one of the fixed wheels is disposed above the rotary inner cylinder to constitute the first deflecting pulley.

4. The foldable small flying and squat rack according to claim 3, wherein a hole plate fixing mechanism for limiting the rotation of the pivot racks is disposed between the two pivot racks and the fixed rack, and a first spring bolt is further disposed on the connecting sheet; and the hole plate fixing mechanism comprises an upper rotary sleeve fixedly connected with the upper part of the fixed rack, and the rotary inner cylinder is rotatably connected with the upper rotary sleeve; and the upper rotary sleeve is fixedly connected with an upper fixed hole plate, the upper fixed hole plate is provided with a plurality of upper insertion holes, and the first spring bolt is inserted into an upper insertion hole of the plurality of upper insertion holes to fix the position of the connecting sheet relative to the upper fixed hole plate; the axes of the upper rotary sleeve and the rotary inner cylinder are both vertical to the ground; and a plurality of upper pivot bearings are provided between the upper rotary sleeve and the rotary inner cylinder, an interior of the rotary inner cylinder is hollow, and the connecting rope passes through the interior of the rotary inner cylinder.

5. The foldable small flying and squat rack according to claim 4, wherein the hole plate fixing mechanism further comprises a lower rotary sleeve fixedly connected with a lower part of the fixed rack, and a rotary column is disposed below the pivot rack; and a plurality of lower pivot bearings are provided between the lower rotary sleeve and the rotary column, and rotating shafts of the upper rotary sleeve and the lower rotary sleeve coincide and are vertical to the ground.

6. The foldable small flying and squat rack according to claim 5, wherein a lower fixed hole plate is fixedly disposed below the lower rotary sleeve, and a plurality of lower insertion holes are disposed on the lower fixed hole plate; and a positioning hole is disposed on the pivot rack, a fixed pin is inserted into a lower insertion hole of the plurality of lower insertion holes, and the fixed pin can be inserted into the positioning hole so as to fix the position of the lower fixed hole plate relative to the pivot rack; and a top of the lower rotary sleeve is provided with a circular cover plate that is fixedly connected with the rotary column by bolts.

7. The foldable small flying and squat rack according to claim 6, wherein each of the two pivot racks is provided with an adjusting wheel set mechanism, and the adjusting wheel set mechanism comprises a crown block and a movable pulley set; and the crown block is rotatably connected with a bottom of the pivot rack, and the movable pulley set is located in the middle of the pivot rack; and the movable pulley set comprises a sliding rack and an upper movable pulley and a lower movable pulley which are rotatably connected with the sliding rack.

8. The foldable small flying and squat rack according to claim 7, wherein the two pivot racks are both provided with flying carriages, and both the two flying carriages are movably connected with the pivot racks; and both the two flying carriages are provided with flying wheel carriers, and two flying guide wheels are rotatably connected in each of the flying wheel carrier; and the flying wheel carrier is rotatably connected with the flying carriage, and one end of the connecting rope is connected to the flying wheel carrier.

9. The foldable small flying and squat rack according to claim 1, wherein a pull-up assembly is disposed at an upper position of the fixed rack, and the pull-up assembly can be fixedly connected with the fixed rack; and the pull-up assembly comprises two fixed plates fixedly connected with the fixed rack, two fixed tubes are fixedly connected between the two fixed plates, and two vertical rods and two inclined rods are fixedly connected between the two fixed tubes.

10. The foldable small flying and squat rack according to claim 1, wherein the counterweight assembly comprises a plurality of weight stacks, and the middle of each weight stack is provided with a yielding hole and an adjusting hole; and an end of the connecting rope close to the counterweight assembly is connected with an adjusting rod, and the adjusting rod is inserted into the yielding hole; and a plurality of matching holes are provided on the adjusting rod, an adjust-

ing pin is inserted into the adjusting hole, and the adjusting pin can pass through both the adjusting hole and the matching holes.

11. The foldable small flying and squat rack according to claim **1**, wherein the fixed rack comprises a chassis, a counterweight slidable bar and a storage plate disposed on the chassis, the counterweight assembly is slidably connected with the counterweight slidable bar, and a plurality of hooks are arranged on the storage plate.

12. A foldable small flying and squat rack, comprising

a fixed rack, pivot racks and pulley blocks disposed on the fixed rack and the pivot racks, wherein the pivot racks comprise two pivot racks that are respectively connected to the fixed rack; and

a counterweight assembly, connected with the fixed rack in a sliding manner, and an upper part of the counterweight assembly is connected with a connecting rope, which winds around the pulley blocks on the fixed rack and the pivot racks in turn to extend out;

wherein each of the two pivot racks is provided with a pivot wheel set mechanism, and a connecting sheet is fixedly disposed on the pivot wheel set mechanism; and a rotary inner cylinder is fixedly disposed on the connecting sheet, the two pivot racks are rotatably connected with the fixed rack respectively through the rotary inner cylinder and rotate to any angle within a limited angle range; and

the connecting rope passes through the rotary inner cylinder, and a pulley block of the pulley blocks comprises a first deflecting pulley and a second deflecting pulley, which are respectively disposed above and below the rotary inner cylinder; and

wherein, a part of the connecting rope located at the position of a rotating axis of the rotary inner cylinder is always in a vertical state, and a tangent point of the connecting rope winding out of the first deflecting pulley and into the second deflecting pulley is located on the axis of the rotary inner cylinder, so as to avoid the connecting rope from being dragged by the rotation of a pivot rack of the pivot racks.

13. The foldable small flying and squat rack according to claim **12**, wherein the pivot wheel set mechanism comprises a pivot wheel rack fixedly connected to the top of the pivot rack, and a pulley block of the pulley blocks comprises a first pivot wheel, a second pivot wheel, a third pivot wheel and a fourth pivot wheel rotatably connected to the pivot wheel rack;

the first pivot wheel is located on a side of the pivot rack far from the fixed rack, and the second pivot wheel, the third pivot wheel and the fourth pivot wheel are located on a side of the pivot rack close to the fixed rack; and the connecting rope passes through the fourth pivot wheel, the third pivot wheel, the second pivot wheel and the first pivot wheel in turn; and

wherein the fourth pivot wheel is disposed below the rotary inner cylinder to constitute the second deflecting pulley.

14. The foldable small flying and squat rack according to claim **13**, wherein fixed wheel carriers are disposed at both sides of an upper part of the fixed rack, and each fixed wheel carrier is rotatably connected with two fixed wheels, wherein one of the fixed wheels is disposed above the rotary inner cylinder to constitute the first deflecting pulley.

15. The foldable small flying and squat rack according to claim **14**, wherein

a hole plate fixing mechanism for limiting the rotation of the pivot racks is disposed between the two pivot racks and the fixed rack, and a first spring bolt is further disposed on the connecting sheet; and the hole plate fixing mechanism comprises an upper rotary sleeve fixedly connected with the upper part of the fixed rack, and the rotary inner cylinder is rotatably connected with the upper rotary sleeve; and the upper rotary sleeve is fixedly connected with an upper fixed hole plate, the upper fixed hole plate is provided with a plurality of upper insertion holes, and the first spring bolt is inserted into an upper insertion hole of the plurality of upper insertion holes to fix the position of the connecting sheet relative to the upper fixed hole plate; the axes of the upper rotary sleeve and the rotary inner cylinder are both vertical to the ground; and a plurality of upper pivot bearings are provided between the upper rotary sleeve and the rotary inner cylinder, an interior of the rotary inner cylinder is hollow, and the connecting rope passes through the interior of the rotary inner cylinder.

16. The foldable small flying and squat rack according to claim **15**, wherein each of the two pivot racks is provided with an adjusting wheel set mechanism, and the adjusting wheel set mechanism comprises a crown block and a movable pulley set; and the crown block is rotatably connected with a bottom of the pivot rack, and the movable pulley set is located in the middle of the pivot rack; and the movable pulley set comprises a sliding rack and an upper movable pulley and a lower movable pulley which are rotatably connected with the sliding rack.

17. The foldable small flying and squat rack according to claim **12**, wherein a pull-up assembly is disposed at an upper position of the fixed rack, and the pull-up assembly can be fixedly connected with the fixed rack; and the pull-up assembly comprises two fixed plates fixedly connected with the fixed rack, two fixed tubes are fixedly connected between the two fixed plates, and two vertical rods and two inclined rods are fixedly connected between the two fixed tubes.

18. A foldable small flying and squat rack, comprising

a fixed rack, pivot racks and pulley blocks disposed on the fixed rack and the pivot racks, wherein the pivot racks comprise two pivot racks that are respectively connected to the fixed rack; and

a counterweight assembly, which is connected with the fixed rack in a sliding manner, and an upper part of the counterweight assembly is connected with a connecting rope, which winds around the pulley blocks on the fixed rack and the pivot racks in turn to extend out;

wherein each of the two pivot racks is provided with a pivot wheel set mechanism, and a connecting sheet is fixedly disposed on the pivot wheel set mechanism; and a rotary inner cylinder is fixedly disposed on the connecting sheet, the two pivot racks are rotatably connected with the fixed rack respectively through the rotary inner cylinder; and

a pulley block of the pulley blocks comprises a first deflecting pulley and a second deflecting pulley; the first deflecting pulley is disposed above the rotary inner cylinder, and the second deflecting pulley is disposed below the rotary inner cylinder;

wherein, the connecting rope winds out of the first deflecting pulley, passes through the rotary inner cylinder and winds into the second deflecting pulley; and the connecting rope in the rotary inner cylinder is always kept in a vertical state, so as to avoid the connecting rope from being dragged by the rotation of the pivot rack.

19. The foldable small flying and squat rack according to claim **18**, wherein a tangent point of the connecting rope winding out of the first deflecting pulley and winding into the second deflecting pulley is on an axis of the rotary inner cylinder.

20. The foldable small flying and squat rack according to claim **18**, wherein the connecting rope between the first deflecting pulley and the second deflecting pulley is parallel or collinear with the axis of the rotary inner cylinder.

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